

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4

Total Marks:40

Annexure A

CASE STUDY

Code of Conduct:

I A G Jayapradhan (192471016) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: A G Jayapradhan

Annexure A

CASE STUDY

Case Study 1 — Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains:

- I. If a patient has Fever AND Cough $\rightarrow$ Possible Flu
- II. If a patient has Fever AND Rash $\rightarrow$ Possible Measles
- III. If a patient has Cough AND Breathlessness $\rightarrow$ Possible Pneumonia
- IV. Facts about Patient A: Fever = True, Cough = True, Breathlessness = True

- (a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.
- (b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.
- (c) The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

Case Study 2 — Autonomous Traffic Management Agent

A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing:

- I. $\forall x: \text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$ [Rule 1]
- II. $\forall x: \text{ClearPath}(x) \rightarrow \text{GreenSignal}(x) \wedge \neg \text{RedSignal}(x)$ [Rule 2]
- III. $\forall x \forall y: \text{Vehicle}(x) \wedge \text{Behind}(x,y) \wedge \text{GreenSignal}(y) \rightarrow \text{Proceed}(x)$ [Rule 3]
- IV. Facts: $\text{Vehicle}(\text{Ambulance}), \text{EmergencyType}(\text{Ambulance}), \text{Vehicle}(\text{CarA}), \text{Behind}(\text{CarA}, \text{Ambulance})$

- (a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set θ (MGU) at each step.
- (b) Using the Resolution method, prove that 'Proceed (CarA)' is true. Convert relevant clauses to CNF and show each resolution step.
- (c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are needed and justify logically.

Case Study 3 — Agricultural Expert Reasoning System

An agricultural advisory logical agent operates with the following propositional knowledge base (KB):

- 1) P1: $\text{SoilDry} \rightarrow \text{IrrigationNeeded}$
- 2) P2: $\text{IrrigationNeeded} \wedge \text{CropWheat} \rightarrow \text{ApplyDripMethod}$
- 3) P3: $\neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$
- 4) Facts: $\text{SoilDry} = \text{True}, \text{CropWheat} = \text{True}$

- (a) Convert all sentences P1, P2, P3 and the facts into Conjunctive Normal Form (CNF). Show each conversion step (eliminate implication $\rightarrow$ move $\neg$ inward $\rightarrow$ distribute).
- (b) Using Resolution Refutation, prove the theorem 'ApplyDripMethod'. Negate the goal, combine with the KB in CNF, and derive the empty clause $\square$. Draw the resolution proof tree.
- (c) If the fact 'SoilDry' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'CropAtRisk' still follow? Apply appropriate reasoning patterns to justify.

Case Study 4 — Wumpus World Knowledge Agent

An AI agent navigating the Wumpus World perceives: Stench at [1,2], Breeze at [1,1], Glitter at [2,2]. The agent uses a propositional KB with these rules:

- 1) $\text{Stench}[1,2] \rightarrow \text{Wumpus is adjacent to cell}[1,2]$
- 2) $\text{Breeze}[1,1] \rightarrow \text{Pit is adjacent to cell}[1,1]$
- 3) $\text{Glitter}[x,y] \rightarrow \text{Gold is at}[x,y]$
- 4) $\neg \text{Wumpus}[x,y] \wedge \neg \text{Pit}[x,y] \rightarrow \text{Safe}[x,y]$

(a) Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.

(b) Using Forward Chaining from the agent's percepts, identify the possible locations of the Wumpus and Pit. Show each reasoning step precisely.

(c) Evaluate whether the path $[1,1] \rightarrow [1,2] \rightarrow [2,2]$ is safe for the agent to collect the gold. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

ANSWER

These are comprehensive AI/Logic case studies. Below are complete solutions with reasoning steps.

Case Study 1 – Smart Medical Diagnosis System

(a) Propositional Logic Representation

Symbols

- F = Patient has Fever
- C = Patient has Cough
- R = Patient has Rash
- B = Patient has Breathlessness
- FI = Patient has Flu
- M = Patient has Measles
- P = Patient has Pneumonia

Rules

1. Fever AND Cough $\rightarrow$ Flu

[	F		$\wedge$	C		$\rightarrow$	FI
]							

2. Fever AND Rash $\rightarrow$ Measles

[	F		$\wedge$	R		$\rightarrow$	M
]							

3. Cough AND Breathlessness $\rightarrow$ Pneumonia

[	C		$\wedge$	B		$\rightarrow$	P
]							

Facts

[	F						
]							
[	C						
]							
[	B						
]							

(b) Forward Chaining

Initial Facts

Working Memory = {F, C, B}

Step 1

Check Rule 1

$F \wedge C \rightarrow FI$

Both F and C are True.

Rule fires.

Derived:

FI

Working Memory

{F, C, B, FI}

Step 2

Check Rule 2

$F \wedge R \rightarrow M$

Rash is absent.

Rule cannot fire.

Step 3

Check Rule 3

$C \wedge B \rightarrow P$

Both facts available.

Rule fires.

Derived

P

Working Memory

{F, C, B, FI, P}

Final Diagnoses

- **Possible Flu**
 - **Possible Pneumonia**
-

(c) Backward Chaining

Goal:

Pneumonia (P)

Search rule producing P.

Rule

$C \wedge B \rightarrow P$

Sub-goals

Need C

Need B

Check KB

C = True

B = True

Both facts found.

Therefore

P = True

Goal Reduction Tree

P (Pneumonia)

|

$C \wedge B \rightarrow P$

$\frac{\quad}{C(\text{True})} \quad \frac{\quad}{B(\text{True})}$

SUCCESS

Case Study 2 – Autonomous Traffic Management Agent

(a) Unification

Rule 1

[
Vehicle(x) ∧ EmergencyType(x) → ClearPath(x)
]

Facts

Vehicle(Ambulance)

EmergencyType(Ambulance)

Step 1

Match

Vehicle(x)

Vehicle(Ambulance)

Substitution

[
θ₁ = {x/Ambulance}
]

Step 2

Apply θ₁

EmergencyType(Ambulance)

Matches fact.

Final MGU

[
θ = {x/Ambulance}
]

Inference

ClearPath(Ambulance)

(b) Resolution Proof

Goal

Proceed(CarA)

Negated Goal

¬Proceed(CarA)

Convert to CNF

Rule 1

¬Vehicle(x) ∨ ¬EmergencyType(x) ∨ ClearPath(x)

Rule 2

¬ClearPath(x) ∨ GreenSignal(x)

¬ClearPath(x) ∨ ¬RedSignal(x)

Rule 3

¬Vehicle(x) ∨ ¬Behind(x,y)

∨ ¬GreenSignal(y)

∨ Proceed(x)

Facts

Vehicle(Ambulance)

EmergencyType(Ambulance)

Vehicle(CarA)

Behind(CarA,Ambulance)

Negated Goal

$\neg$ Proceed(CarA)

Resolution Steps

From Rule 1 + Facts

Vehicle(Ambulance)

EmergencyType(Ambulance)

$\Rightarrow$ ClearPath(Ambulance)

Rule 2

ClearPath(Ambulance)

$\Rightarrow$ GreenSignal(Ambulance)

Rule 3

Substitute

$x = \text{CarA}$

$y = \text{Ambulance}$

Clause becomes

$\neg \text{Vehicle}(\text{CarA})$

$\vee \neg \text{Behind}(\text{CarA}, \text{Ambulance})$

$\vee \neg \text{GreenSignal}(\text{Ambulance})$

$\vee \text{Proceed}(\text{CarA})$

Resolve with

Vehicle(CarA)

$\downarrow$

$\neg \text{Behind}(\text{CarA}, \text{Ambulance})$

$\vee \neg \text{GreenSignal}(\text{Ambulance})$

$\vee \text{Proceed}(\text{CarA})$

Resolve with

Behind(CarA,Ambulance)

$\downarrow$

$\neg \text{GreenSignal}(\text{Ambulance})$

$\vee \text{Proceed}(\text{CarA})$

Resolve with

GreenSignal(Ambulance)

↓

Proceed(CarA)

Resolve with

$\neg$ Proceed(CarA)

↓

□

Empty clause proved.

Hence

Proceed(CarA) is TRUE.

(c) Analysis

Current KB only supports one emergency vehicle at a time.

Problems:

- **No priority between two ambulances.**
- **Signal conflicts not handled.**
- **Route conflicts ignored.**

Additional Rules

1.

$\text{EmergencyPriority}(x) \wedge \text{HigherPriority}(x,y)$

→ $\text{GreenSignal}(x)$

2.

$\text{ClearPath}(x) \wedge \text{Intersects}(x,y)$

→ $\text{Wait}(y)$

3.

$\text{EmergencyType}(x)$

→ $\text{HighestPriority}(x)$

These rules enable simultaneous emergency management while preventing collisions.

Case Study 3 – Agricultural Expert Reasoning System

(a) CNF Conversion

P1

Original

$\text{SoilDry} \rightarrow \text{IrrigationNeeded}$

Remove implication

$\neg \text{SoilDry} \vee \text{IrrigationNeeded}$

CNF

$\neg \text{SoilDry} \vee \text{IrrigationNeeded}$

P2

Original

$\text{IrrigationNeeded} \wedge \text{CropWheat}$

→ ApplyDripMethod

Implication

$\neg(\text{IrrigationNeeded} \wedge \text{CropWheat})$

$\vee \text{ApplyDripMethod}$

De Morgan

$\neg \text{IrrigationNeeded}$

$\vee \neg \text{CropWheat}$

$\vee \text{ApplyDripMethod}$

CNF

$\neg \text{IrrigationNeeded}$

$\vee \neg \text{CropWheat}$

$\vee \text{ApplyDripMethod}$

P3

Original

$\neg \text{ApplyDripMethod} \wedge \text{CropWheat}$

$\rightarrow \text{CropAtRisk}$

Implication

$\neg(\neg \text{ApplyDripMethod} \wedge \text{CropWheat})$

$\vee \text{CropAtRisk}$

De Morgan

ApplyDripMethod

$\vee \neg \text{CropWheat}$

$\vee \text{CropAtRisk}$

CNF

ApplyDripMethod

$\vee \neg \text{CropWheat}$

$\vee \text{CropAtRisk}$

Facts

SoilDry

CropWheat

(b) Resolution Refutation

Goal

ApplyDripMethod

Negation

$\neg \text{ApplyDripMethod}$

Clauses

1. $\neg \text{SoilDry} \vee \text{IrrigationNeeded}$

2. $\neg \text{IrrigationNeeded}$

$\vee \neg \text{CropWheat}$

$\vee \text{ApplyDripMethod}$

3. SoilDry

4. CropWheat

5. $\neg$ ApplyDripMethod

Resolution

3 + 1

↓

IrrigationNeeded

IrrigationNeeded + Clause 2

↓

$\neg$ CropWheat

$\vee$ ApplyDripMethod

CropWheat + previous

↓

ApplyDripMethod

ApplyDripMethod + Negated Goal

↓

□

Goal proved.

Resolution Tree



(c) Removing SoilDry

Remaining facts

CropWheat

Cannot derive

IrrigationNeeded

Therefore

ApplyDripMethod

cannot be inferred.

Rule P3

$\neg$ ApplyDripMethod $\wedge$ CropWheat

→ CropAtRisk

does not imply $\neg$ ApplyDripMethod; it only states what follows if $\neg$ ApplyDripMethod is known. Since $\neg$ ApplyDripMethod cannot be proven from the remaining KB, CropAtRisk also cannot be concluded.

Conclusion:

- IrrigationNeeded ✗ Not derived
- ApplyDripMethod ✗ Not derived
- CropAtRisk ✗ Not derived

Case Study 4 – Wumpus World Knowledge Agent

(a) Belief State

Percepts

Stench(1,2)

Breeze(1,1)

Glitter(2,2)

Apply Modus Ponens

Rule 1

Stench(1,2)

→ Wumpus adjacent to (1,2)

Derived

AdjacentWumpus(1,2)

Rule 2

Breeze(1,1)

→ Pit adjacent to (1,1)

Derived

AdjacentPit(1,1)

Rule 3

Glitter(2,2)

→ Gold(2,2)

Derived

Gold(2,2)

Belief State

Stench(1,2)

Breeze(1,1)

Glitter(2,2)

AdjacentWumpus(1,2)

AdjacentPit(1,1)

Gold(2,2)

(b) Forward Chaining

Step 1

Stench

↓

Possible Wumpus cells

(1,1)

(1,3)

(2,2)

Step 2

Breeze

↓

Possible Pit cells

(1,2)

(2,1)

Step 3

Glitter

↓

Gold

(2,2)

Final Inference

Possible Wumpus

(1,1)

(1,3)

(2,2)

Possible Pit

(1,2)

(2,1)

Gold

(2,2)

(c) Safety Evaluation

Proposed Path

(1,1)

↓

(1,2)

↓

(2,2)

Analysis:

- Breeze at (1,1) indicates there may be a Pit adjacent to (1,1), making (1,2) a possible pit location.
- Stench at (1,2) indicates there may be a Wumpus adjacent to (1,2), making (2,2) a possible Wumpus location.
- Glitter confirms Gold at (2,2), but it does not guarantee the square is safe.
- Rule 4 ($\neg \text{Wumpus}[x,y] \wedge \neg \text{Pit}[x,y] \rightarrow \text{Safe}[x,y]$) cannot be applied because neither $\neg \text{Wumpus}$ nor $\neg \text{Pit}$ has been established for the relevant cells.

Conclusion:

The path $(1,1) \rightarrow (1,2) \rightarrow (2,2)$ cannot be declared safe based on the current knowledge base. The agent requires additional percepts or logical evidence to prove that (1,2) and (2,2) are free of pits and the Wumpus before moving.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding ; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand